MADHU S

Bengaluru, Karnataka | +91 97405 95939 | madhu.s.3606@gmail.com
github.com/Madhu-36 | linkedin.com/in/madhu-s- | Portfolio: madhu-36.github.io/My_Portfolio

Seeking: Internship / Entry-Level role in AI & ML, Cybersecurity , Game or Software Development

SUMMARY

Third-year B.E. (AI & Data Science) student who builds and secures real systems. Built an AI-driven adaptive firewall (LSTM-Autoencoder + reinforcement learning with eBPF/XDP), a local voice AI assistant, and full-stack web apps. Completed two ethical hacking internships (Burp Suite, Nmap, OWASP Top 10) and led teams at Smart India Hackathon 2025 and NASA Space Apps Challenge 2025.

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript, TypeScript, Rust, C, Dart
AI / ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, Ollama / Local LLMs (Llama 3.1, Whisper)
Cybersecurity: Kali Linux, Burp Suite, Nmap, Metasploit, Wireshark, OWASP Top 10 (SQLi, XSS, IDOR), Penetration Testing
Web & Databases: Django, Node.js, React, Bootstrap, HTML5, CSS3, MySQL, MongoDB
Tools & Cloud: Git, Linux CLI, Docker, AWS Cloud Foundations, Unity 6.0 (C#)

INTERNSHIPS

Ethical Hacker Intern | ACMEGRADE *Dec 2025 – Jun 2026*

- Ran web and network vulnerability assessments with Burp Suite and Nmap, finding XSS, SQL injection, IDOR and access-control flaws.
- Manually verified intercepted HTTP traffic to remove false positives and prioritise the vulnerabilities that mattered.
- Wrote step-by-step Proof-of-Concepts and remediation reports; retested patches with senior engineers to confirm fixes.

Ethical Hacker Intern | Jutrm Career Academy *Mar 2026*

- Performed hands-on penetration tests and vulnerability assessments on real-time network infrastructure projects.
- Analysed system vulnerabilities and kernel-level security requirements to reduce attack surface; earned Certificate of Completion.

PROJECTS

Next-Gen Adaptive Firewall (AI Security) | Python, PyTorch, eBPF/XDP, Scapy | GitHub *Ongoing*

- Real-time network anomaly detection using an LSTM-Autoencoder, with a PPO reinforcement-learning policy deciding responses.
- Automates kernel-level filtering rules via eBPF/XDP to block DDoS and zero-day threats without human intervention.

MXB – Local Voice AI Assistant | Python, Llama 3.1, Whisper, Ollama | GitHub

- Offline voice assistant for Windows running a local LLM, with TF-IDF semantic memory and automated task workflows.

CodeShield – Security Layer for AI Dev Assistants | Python | GitHub

- Safe execution sandbox plus AST-based vulnerability scanning for code generated by AI assistants.

Game Complex Management System | Python, Django, MySQL | GitHub *Mar 2025 – Jun 2025*

- Full-stack web app with real-time booking, inventory tracking and member management, backed by custom REST-style APIs.

Also on GitHub: DevIgnite (Rust developer-environment manager), Packet-way (Node.js packet-inspection & network forensics engine), interactive 3D portfolio (React, Three.js).

LEADERSHIP & ACHIEVEMENTS

- **Team Leader, Smart India Hackathon 2025** – led institute team "EW Root Coders" building a digital solution for a real-world problem statement.
- **Team Leader, NASA Space Apps Challenge 2025** – led project planning, strategy and technical execution.
- **First Place, District Mathematical Competition** – ranked 1st for problem-solving speed and accuracy.

EDUCATION

B.E. Artificial Intelligence & Data Science | East West Institute of Technology (EWIT), Bengaluru *2023 – 2027 (Expected)*

12th Grade (State Board) | KLE Independent PU College, Bengaluru *2022 – 2023*

10th Grade (ICSE) | Elite Public School, Bengaluru *2020 – 2021*

CERTIFICATIONS & TRAINING

- **AWS Academy Graduate – AWS Cloud Foundations** (Aug 2026)
- **Unity Junior Programmer Pathway** (Jul 2026)
- **Forage Job Simulations:** Cybersecurity Operations – DataCom (Jul 2026); Cyber – Deloitte Australia (Apr 2026); Security Analyst – Tata (Apr 2026); Cloud Architecture – Amazon
- **Courses:** Data Science using Python, Advanced Java, Web Technologies, Unity 5.x & 6.0 Game Development